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The solubility, thermal stability and the basic character of the hydroxides increase down the group. The alkaline earth metal hydroxides are, however, less basic and less stable than alkali metal hydroxides. Beryllium hydroxide is amphoteric in nature as it reacts with both acid and alkali.

$$Be(OH)_2 + 2\ NaOH \rightarrow Na_2BeO_2 + 2H_2O$$

$$Be(OH)_2 + 2HCl \rightarrow BeCl_2 + 2H_2O$$

c) Halides:

Alkaline earth metals form halides with general formula MX_2. They can be prepared by heating metals with halogens on heating.

$$M + X_2 \longrightarrow MX_2$$

Beryllium halides are covalent on account of smaller size of Be^{+2}. Beryllium halides are hygroscopic, fume in moist air and soluble in organic solvents. Beryllium chloride has a chain structure in the solid state as shown in figure 5.9 (structure-a). In the vapour phase $BeCl_2$ tends to form a chloro-bridged dimer (structure-c) which dissociates into the linear monomer at high temperatures of the order of 1200 K. (structure-b).

Except beryllium halides, all the other halides of alkaline earth metals are ionic in nature. Chloride and fluorides of the other metals are ionic solids. These are good conductors of electricity in fused state and in aqueous solutions. The tendency to form halide hydrates gradually decreases (for example, $MgCl_2.8H_2O$, $CaCl_2.6H_2O$, $SrCl_2.6H_2O$ and $BaCl_2.2H_2O$) down the group.

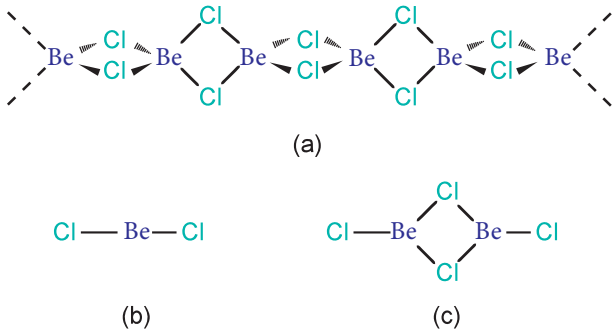

Figure 5.9 Structure of beryllium chloride

Salts of oxo acids

The alkaline earth metals form salts of oxo acids. Some of these are given below:

Carbonates:

All the carbonates decompose on heating to give carbon dioxide and the oxide.

$$MCO_3 \xrightarrow{\Delta} MO + CO_2$$

- The solubility of carbonates in water decreases down the group.
- The thermal stability increases down the group with increasing cationic size.

Table 5.13 Decomposition temperature of alkaline metal carbonates and sulphates

Element	Decomposition temp for carbonates (in °C)	Decomposition temp for sulphates (in °C)
Be	25	500
Mg	540	895
Ca	900	1149
Sr	1290	1374
Ba	1360	-